AdventureWorks Analytics

Data-Driven Retail Sales Optimization

Raw Excel → Python ETL → MySQL Star Schema → Power BI

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1. Project Objectives & Business Goals

Business Problem

Adventure Works operates through Internet (B2C) and Reseller (B2B) channels but lacks a centralized analytical system. Data is fragmented across POS, CRM, eCommerce, and inventory systems. Brian Welcker, VP of Sales, cannot answer territorial, customer, or profitability questions without manual consolidation.

Business Goal

Build a centralized analytical pipeline integrating all sales data into a Star Schema warehouse, enabling accurate reporting on performance, customer segments, and profitability; delivered as a five-page interactive Power BI dashboard accessible to non-technical decision makers.

Objectives → Deliverables Mapping

Case Study Objective	Delivered By
Identify underperforming sales territories	Page 2: Territory heatmap (vw_territory_performance)
Analyse revenue and profitability trends	Page 1: YoY trend + Decomposition Tree + Q1 SQL
Customer segmentation by occupation and income	Page 3: RFM + Occupation chart + CustomerSegment donut
Target right age group for sales growth	Page 3: Internet Revenue by AgeGroup
Compare Internet vs Reseller performance	Page 5: Channel margin comparison + Q4 SQL

KPI Register

KPI	Value	Type	Source / Page
Total Revenue	$109.8M	Required	SUM both channels / P1
Internet Revenue	$29.4M (26.6%)	Required	SUM Internet Sales / P1, P5
Reseller Revenue	$80.5M (73.4%)	Required	SUM Reseller Sales / P1, P5
Total Profit	$12.6M	Required	SUM Profit both channels / P1

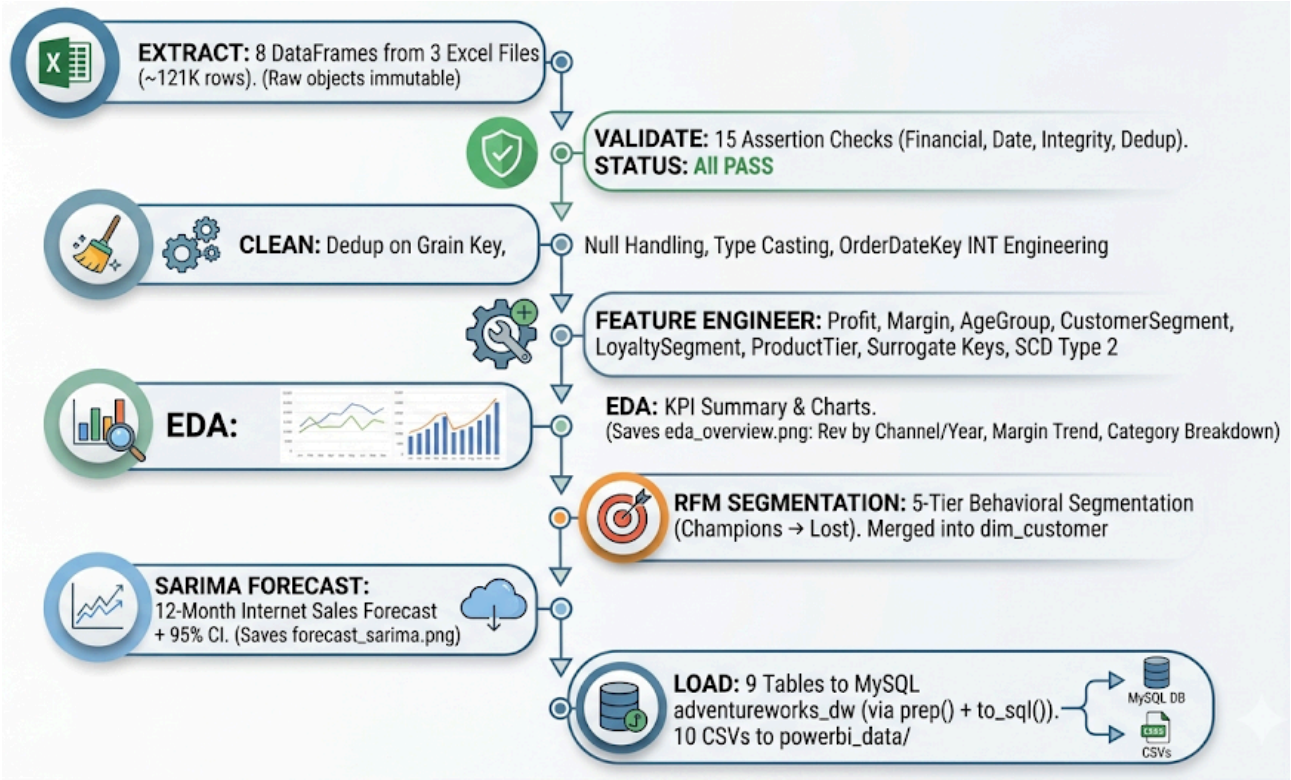
Profit Margin %	11.43%	Required	DIVIDE(Profit, Revenue) / P1
Total Orders	31,455	Required	DISTINCTCOUNT SalesOrderNumber / P1
Unique Customers	18,484	Required	DISTINCTCOUNT CustomerKey (Internet) / P3
Revenue by Territory	SW $24.2M / AUS $10.7M	Required	vw_territory_performance / P2
Revenue per Customer	$1,588	Required	Internet Revenue / Unique Customers / P3
Customer Segments (RFM)	Champions 2.1K | Lost 1.9K	Required	Python RFM / P3
YoY Revenue Growth	2012: +34.7% | 2013: +46.7%	Required	DATEADD time intelligence / P1
Internet Margin %	41.15% → green	Added – channel efficiency	DIVIDE(Internet Profit, Internet Revenue) / P5
Reseller Margin %	0.58% → red	Added – channel efficiency	DIVIDE(Reseller Profit, Reseller Revenue) / P5
Territory Margin %	Australia 33.6% (highest)	Added – geographic profitability	Combined profit/revenue by territory / P2

Low Margin Territories	5 below 10%	Added – risk flagging	COUNTROWS + FILTER measure / P2
Avg Order Value	$3,491	Added – basket size	Total Revenue / Total Orders / P1
Internet Revenue Target	$35M target vs $29.4M	Added – goal tracking	Gauge visual / P1
Product Revenue Rank	DENSE_RANK() per product	Added – portfolio analysis	Window function in SQL view / P4

Note: Why additional KPIs were created

The case study specifies core KPIs (Revenue, Margin %, Territory, Customer Segment, YoY Growth). Additional KPIs were derived during EDA because they surfaced analytically significant patterns not visible from the required set alone. Internet Margin % vs Reseller Margin % is the single most important finding in the project; it was not possible to surface this from total blended margin. Territory Margin %, Low Margin Territories, and Product Revenue Rank were similarly added because they answered questions the required KPIs could not.

2. Python ETL Pipeline



Assumptions

(1) Null DiscountAmount treated as 0; a null discount is economically equivalent to no discount.

(2) YearlyIncome nulls filled with median, not mean; median is robust to income outliers.

(3) UnitPrice = SalesAmount / SalesOrderLineNumber is a proxy; no OrderQuantity column exists in source data.

(4) Age and tenure calculated against fixed reference date 2014-12-31 for reproducibility.

(5) All customer analysis is Internet Sales only; FactResellerSales has no CustomerKey column.

(6) 60% COGS assumption not used; TotalProductCost from source data is used directly.

Step 1: Extract

```
FACT_INTERNET = 'FactInternetSales.xlsx'
FACT_RESELLER = 'FactResellerSales.xlsx'
DIM_TABLES    = 'DimTables.xlsx'
fact_internet = pd.read_excel(FACT_INTERNET)
fact_reseller = pd.read_excel(FACT_RESELLER)
dim_customer  = pd.read_excel(DIM_TABLES, sheet_name='DimCustomer')
dim_product   = pd.read_excel(DIM_TABLES, sheet_name='DimProduct')
dim_reseller  = pd.read_excel(DIM_TABLES, sheet_name='DimReseller')
dim_geography = pd.read_excel(DIM_TABLES, sheet_name='DimGeography')
dim_territory = pd.read_excel(DIM_TABLES, sheet_name='DimSalesTerritory')
dim_employee  = pd.read_excel(DIM_TABLES, sheet_name='DimEmployee')
```

```
Extraction complete
  FactInternetSales        :  60,398 rows  13 cols
  FactResellerSales        :  60,899 rows  14 cols
  DimCustomer              :  18,484 rows  13 cols
  DimProduct               :     397 rows  7 cols
  DimReseller              :     701 rows  4 cols
  DimGeography             :     655 rows  6 cols
  DimSalesTerritory        :      11 rows  4 cols
  DimEmployee              :      21 rows  9 cols
```

Step 2: Validate (15 Assertion Checks)

The validation gate runs before any cleaning or transformation. If a referential integrity check fails, it points directly at the source file; not at something the pipeline may have introduced. This is the standard production data quality pattern.

```
VALIDATION REPORT   PASS: 15  FAIL: 0  of 15 checks
                                  Check Status                      Detail
           FIS: No negative SalesAmount   PASS                    min=2.29
           FIS: No negative ProductCost   PASS                    min=0.86
              FIS: No future OrderDates   PASS              max=2014-01-28
                FIS: OrderDate not null   PASS                     nulls=0
             FIS: ShipDate >= OrderDate   PASS Ship must not precede Order
     FIS: All ProductKeys in DimProduct   PASS                   orphans=0
   FIS: All CustomerKeys in DimCustomer   PASS                   orphans=0
        FIS: No duplicate OrderLine keys   PASS                     dupes=0
           FRS: No negative SalesAmount   PASS
              FRS: No future OrderDates   PASS              max=2013-11-29
   FRS: All ResellerKeys in DimReseller   PASS
  DimCustomer: No duplicate CustomerKey   PASS
    DimProduct: No duplicate ProductKey   PASS
     DimCustomer: YearlyIncome positive   PASS
DimGeography: No duplicate GeographyKey   PASS

All checks passed - safe to proceed.
```

Step 3: Clean & Feature Engineering

Business question: What new analytical value is being created and why?

Cleaning is minimal because the source data passed all 15 validation checks. The one consequential engineering decision in cleaning is the OrderDateKey: an integer (YYYYMMDD format) that makes fact-to-date joins use index equality instead of DATE_FORMAT() function calls, which cannot use indexes (non-sargable). At 120,000 fact rows this is measurably faster in every SQL query and every time-intelligence DAX measure.

Feature engineering creates the analytical columns that do not exist anywhere in the source files. These become the KPI-powering fields in the warehouse:

```
# Fact Table Features
for df, name in [(fi,'Internet'),(fr,'Reseller')]:
    df['Revenue']       = df['SalesAmount']
    df['Profit']        = df['SalesAmount'] - df['TotalProductCost']
    df['ProfitMargin']  = (df['Profit'] / df['SalesAmount'].replace(0,np.nan)*100).round(2)
    df['DiscountRate']  = (df['DiscountAmount'] / df['SalesAmount'].replace(0,np.nan)*100).round(2).fillna(0)
    # UnitPrice = SalesAmount / SalesOrderLineNumber (line quantity proxy)
    df['UnitPrice']     = (df['SalesAmount'] / df['SalesOrderLineNumber'].replace(0,np.nan)).round(2)
    df['OrderYear']     = df['OrderDate'].dt.year
    df['OrderMonth']    = df['OrderDate'].dt.month
    df['OrderQuarter']  = df['OrderDate'].dt.quarter
    df['DaysToShip']    = (df['ShipDate'] - df['OrderDate']).dt.days
    df['DaysToDue']     = (df['DueDate']  - df['OrderDate']).dt.days
    print(f'{name} features added')

fr['ResellerMargin'] = (fr['Profit'] / fr['TotalProductCost'].replace(0,np.nan)*100).round(2)
```

```
# DimCustomer Features
REF = pd.Timestamp('2014-12-31')
dc['Age'] = ((REF - dc['BirthDate']).dt.days / 365.25).fillna(0).astype(int).clip(lower=0)

dc['AgeGroup'] = pd.cut(dc['Age'], bins=[0,25,35,45,55,150],
                        labels=['18-25','26-35','36-45','46-55','55+'])

dc['IncomeBand'] = pd.cut(dc['YearlyIncome'],
                          bins=[0,40000,80000,120000,1e9],
                          labels=['Low (<40K)','Mid (40K-80K)','High (80K-120K)','Very High (120K+)'])

dc['CustomerSegment'] = pd.cut(dc['YearlyIncome'],
                          bins=[0,40000,80000,120000,1e9],
                          labels=['Budget','Standard','Premium','Luxury'])

dc['TenureYears']    = ((REF - dc['FirstPurchaseDate']).dt.days / 365.25).round(1)
dc['LoyaltySegment'] = pd.cut(dc['TenureYears'],
                          bins=[-1,2,5,100],
                          labels=['New (<2 Yr)','Regular (2-5 Yr)','Loyal (5+ Yr)'])
```

→ Internet features added

→ Reseller features added

> **Key Insight**
>
> The 36-45 age group and Professional occupation are not coincidentally the top segments; these features were engineered precisely to answer the case study objectives about age-group targeting and occupation-based spending. Without AgeGroup and CustomerSegment columns, these questions cannot be answered from the raw data.

Business Action

These engineered segments are used in all Power BI slicers on Page 3. A business user can filter by AgeGroup or CustomerSegment without any understanding of how those columns were created.

Surrogate Keys + SCD Type 2

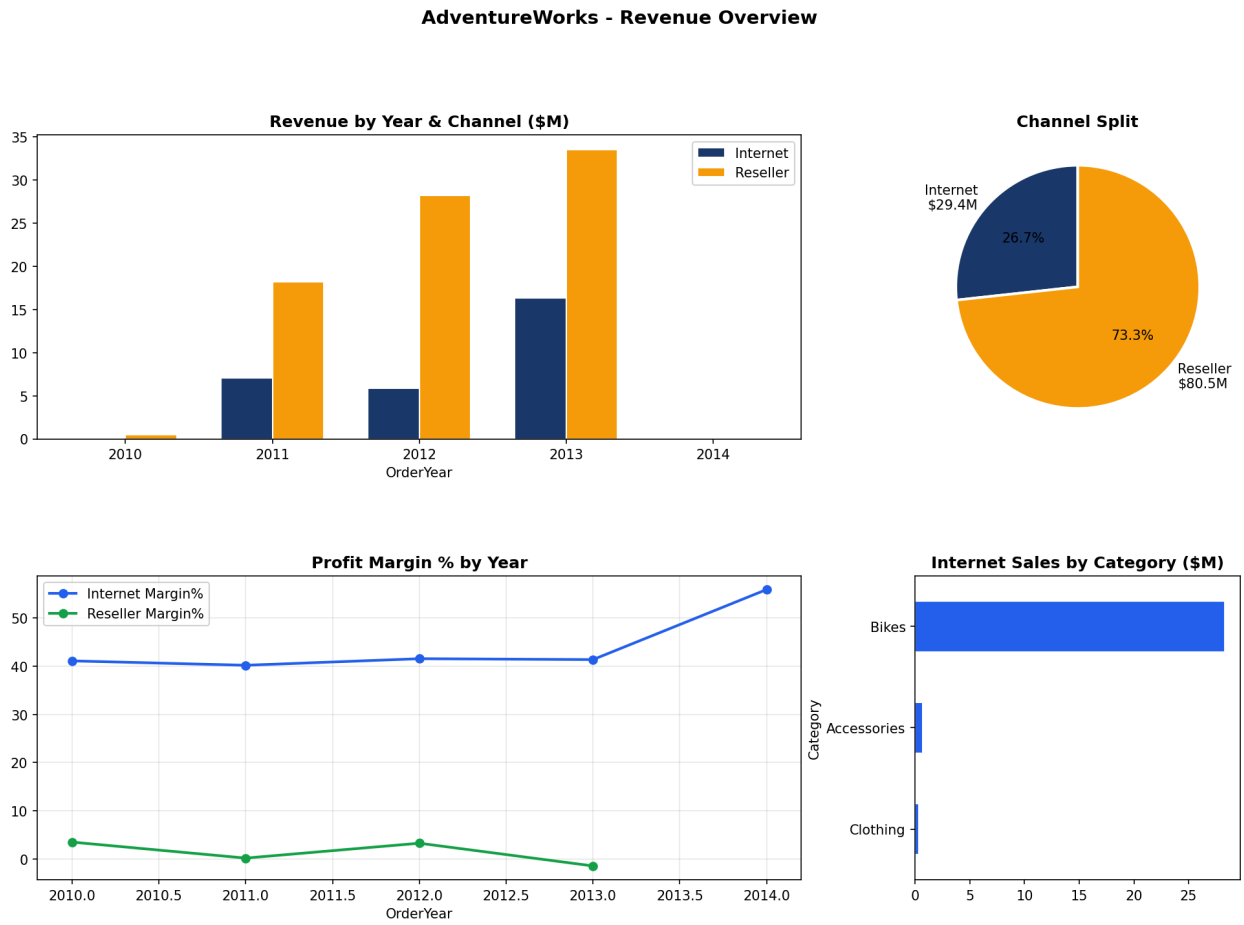
```
Surrogate keys assigned:
   CustomerSK: 1-18484  ProductSK: 1-397  ResellerSK: 1-701
SCD Type 2 fields added to dim_customer and dim_product
   Fields: EffectiveDate | EndDate (9999-12-31 = active) | IsCurrent | RowVersion

How a delta update works (SCD2 logic):
  STEP 1 - Identify changed row by comparing source vs warehouse hash
  STEP 2 - UPDATE old row: SET EndDate=today(), IsCurrent=0
  STEP 3 - INSERT new row: new SK, EffectiveDate=today(), EndDate=9999-12-31, IsCurrent=1
  RESULT - Full history preserved, current row flagged with IsCurrent=1
```

Step 4: Exploratory Data Analysis

Business question: What does the data tell us before we build anything?

EDA runs on the processed data; after cleaning and feature engineering, the results reflect the actual data that will be loaded into the warehouse. The KPI summary output below cross-validates every Power BI card value before the warehouse is even built.

```
  ADVENTUREWORKS - KPI SUMMARY
Total Revenue       : $    109,809,274
Internet Revenue    : $     29,358,677  (26.7%)
Reseller Revenue    : $     80,450,597  (73.3%)
Total Profit        : $     12,551,366
Profit Margin       : 11.4%
Internet Orders     :          27,659
Reseller Orders     :           3,796
Unique Customers    :          18,484
Unique Products     :             397
```



> **Key Insight**
>
> EDA immediately revealed the core finding: Reseller dominates revenue (73.3%) while Internet carries higher margin. This single observation shaped the entire dashboard structure; five pages are designed to answer why this gap exists and what the business should do about it.

Step 5: RFM Customer Segmentation

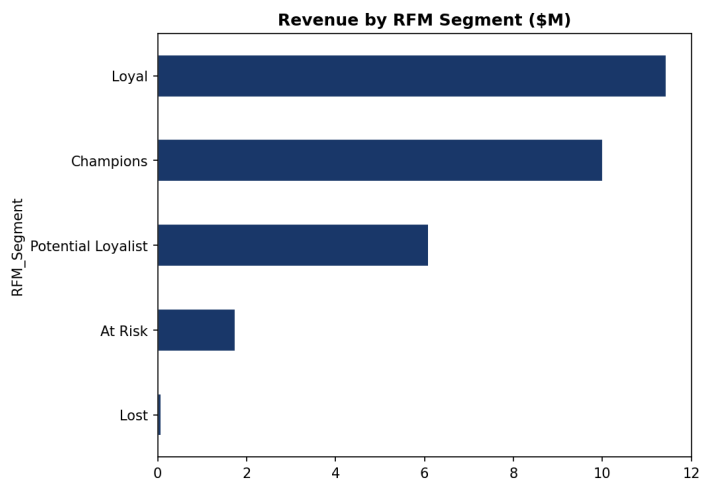
Business question: Which customers drive the most revenue, and which are at risk of churn?

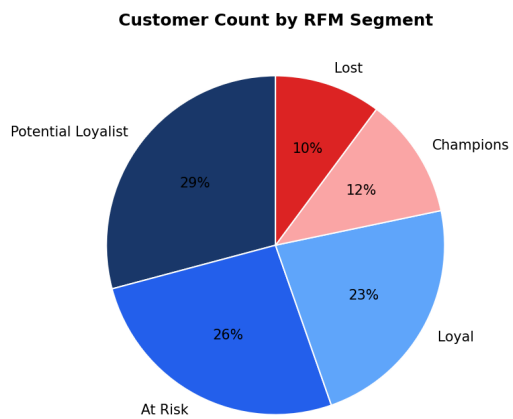
RFM compresses each customer's entire transaction history into three numbers. Recency: days since their last order (lower = better customer). Frequency: how many

unique orders they placed. Monetary: their total spend. Each is scored 1-4 via quantile bins, combined into RFM_Score (range 3-12), and mapped to five named segments. The result is merged back into dim_customer so the segment becomes a Power BI dimension attribute.

```python
ADATE = fi['OrderDate'].max() + pd.Timedelta(days=1)

rfm = (fi.groupby('CustomerKey')
        .agg(Recency  =('OrderDate',   lambda d:(ADATE-d.max()).days),
             Frequency=('SalesOrderNumber','nunique'),
             Monetary =('SalesAmount','sum'))
        .reset_index())

rfm['R_Score'] = pd.qcut(rfm['Recency'],   q=4, labels=[4,3,2,1]).astype(int)
rfm['F_Score'] = pd.qcut(rfm['Frequency'].rank(method='first'), q=4, labels=[1,2,3,4]).astype(int)
rfm['M_Score'] = pd.qcut(rfm['Monetary'],  q=4, labels=[1,2,3,4]).astype(int)
rfm['RFM_Score']= rfm['R_Score'] + rfm['F_Score'] + rfm['M_Score']

rfm['RFM_Segment'] = pd.cut(rfm['RFM_Score'], bins=[0,4,6,8,10,12],
                    labels=['Lost','At Risk','Potential Loyalist','Loyal','Champions'])

dc = dc.merge(rfm[['CustomerKey','Recency','Frequency','Monetary',
                   'RFM_Score','RFM_Segment']], on='CustomerKey', how='left')
```

```
RFM complete
RFM_Segment
Potential Loyalist    5387
At Risk               4846
Loyal                 4236
Champions             2133
Lost                  1882
Name: count, dtype: int64

Revenue by segment:
RFM_Segment
Loyal                 $11,435,907
Champions             $10,006,128
Potential Loyalist     $6,098,594
At Risk                $1,742,423
Lost                      $75,625
Name: Monetary, dtype: object
```





Key Insight

Champions are only 11.5% of customers but generate 34% of Internet revenue ($10.0M). The Lost segment is 10.2% of customers but generates only $0.1M. The RFM combo chart on Page 3 makes this disproportionate value visible at a glance; one bar is 100× taller than the other for segments of similar size.

Business Action

Activate a VIP loyalty programme for Champions (2,133 customers). Re-engage At Risk customers (4,846 customers at ~60 days recency) with personalised win-back campaigns. A 20% response rate on At Risk customers recovers approximately $340K in near-lost revenue.

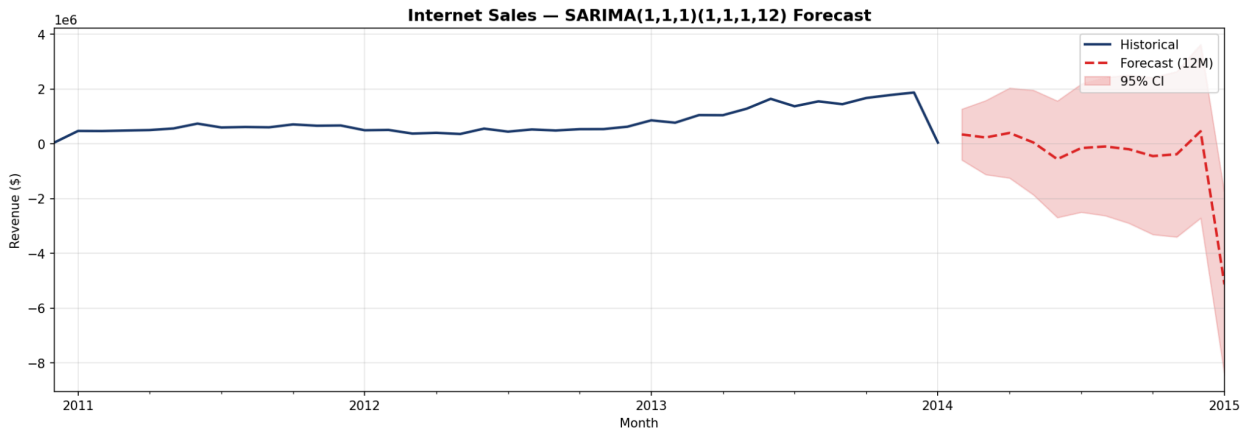
Step 6: SARIMA Revenue Forecast

Business question: What does Internet Sales revenue look like over the next 12 months?

Three forecasting models were evaluated before SARIMA was selected.

- **Simple ETS/Moving Average** [Rejected]: No autocorrelation capture. Produces flat or linear forecasts. Adds no value over a simple average.
- **Facebook Prophet** [Rejected]: Requires 2+ years for reliable seasonal decomposition. On 3 years of monthly data it would overfit the seasonal component.
- **SARIMA(1,1,1)(1,1,1,12) [Selected]:** Designed for short univariate series. AR(1) captures recent trends, I(1) removes drift, MA(1) smooths noise, seasonal (1,1,1,12) captures the 12-month bicycle sales cycle. Native 95% CI bounds.

```
monthly = (fi.groupby(fi['OrderDate'].dt.to_period('M'))['SalesAmount']
            .sum().reset_index())
monthly.columns = ['Month','Revenue']
monthly['Month'] = monthly['Month'].dt.to_timestamp()
monthly = monthly.sort_values('Month').set_index('Month')

try:
    from statsmodels.tsa.statespace.sarimax import SARIMAX
    model   = SARIMAX(monthly['Revenue'],order=(1,1,1),seasonal_order=(1,1,1,12),
            enforce_stationarity=False,enforce_invertibility=False)
    results = model.fit(disp=False)
    fc = results.get_forecast(steps=12)
    fc_mean = fc.predicted_mean; fc_ci = fc.conf_int()
```



Model Limitation

SARIMA requires 24+ months for reliable seasonal decomposition. This model is trained on 36 monthly data points. The forecast is a directional planning indicator only. The intentionally wide 95% CI band in the dashboard honestly reflects this uncertainty.

Key Insight

The SARIMA model projects stable growth continuing into 2014. The 95% CI band widens significantly beyond 3 months, correctly communicating that short-range forecasting (1-3 months) is more reliable than long-range. The forecast validates that the 2013 growth trajectory is structural, not a one-year anomaly.

Business Action

Pre-position inventory for approximately $500K–$600K monthly Internet Sales in Q2 2014. The CI lower bound ($420K) is the conservative planning figure; the CI upper bound ($582K) is the optimistic scenario.

3. MySQL Data Warehouse

Schema Design

A Star Schema was chosen over Snowflake. At this volume (121K rows, 9 tables), additional normalisation adds JOIN hops without storage savings. The denormalised star produces simpler DAX in Power BI and cleaner SQL in analytical views. A staging layer in Python (the prep() function) handles all type conversion before MySQL load. MySQL only receives warehouse-ready data.

```
dim_date                          2,191 rows
dim_customer                     18,484 rows
dim_product                         397 rows
dim_reseller                        701 rows
dim_geography                       655 rows
dim_sales_territory                  11 rows
dim_employee                         21 rows
fact_internet_sales              60,398 rows
fact_reseller_sales              60,855 rows

All tables loaded!
```

> **Note: SCD Type 2 on dim_customer and dim_product**
>
> SCD Type 2 tracks historical attribute changes. If a customer changes occupation from Clerical to Professional, the old row is closed (EndDate = change date, IsCurrent = 0) and a new row is inserted (new SK, EffectiveDate = today, IsCurrent = 1, RowVersion + 1). Power BI always filters WHERE IsCurrent = 1 to see current state. Full history is preserved for retrospective analysis. This is the production data warehouse standard for slowly changing attributes.

Idempotent Index Engineering

Running the index script twice on an already-loaded database throws three different errors. Stored procedures that check information_schema before every DROP make the entire script safe on any run number.

```sql
CREATE PROCEDURE drop_index_if_exists(
    IN tbl  VARCHAR(100),
    IN idx  VARCHAR(100)
)
BEGIN
    IF EXISTS (
        SELECT 1 FROM information_schema.statistics
        WHERE  table_schema = DATABASE()
        AND    table_name   = tbl
        AND    index_name   = idx
    ) THEN
        SET @sql = CONCAT('ALTER TABLE `', tbl, '` DROP INDEX `', idx, '`');
        PREPARE stmt FROM @sql;
        EXECUTE stmt;
        DEALLOCATE PREPARE stmt;
    END IF;
END$$
DELIMITER ;
```

SQL View Definitions

The SQL layer pre-computes joins and aggregations that would require complex multi-table DAX in Power BI. Three views are loaded directly as Power BI visual data sources. All views demonstrate JOIN, GROUP BY, CASE WHEN, window functions, and CTEs.

vw_territory_performance: Page 2 heatmap table

Combines both fact tables using UNION ALL inside a subquery, then pivots Internet and Reseller contributions into separate columns using conditional CASE WHEN aggregation. Demonstrates: multi-table JOIN, UNION ALL, CASE WHEN pivot, NULLIF for safe division.

```sql
DROP VIEW IF EXISTS vw_territory_performance;
CREATE VIEW vw_territory_performance AS
SELECT
    t.SalesTerritoryKey,
    t.SalesTerritoryRegion   AS Territory,
    t.SalesTerritoryCountry  AS Country,
    t.SalesTerritoryGroup    AS Region,
    SUM(CASE WHEN src='Internet' THEN Revenue ELSE 0 END)  AS InternetRevenue,
    SUM(CASE WHEN src='Reseller' THEN Revenue ELSE 0 END)  AS ResellerRevenue,
    SUM(Revenue)                                            AS TotalRevenue,
    SUM(Profit)                                             AS TotalProfit,
    ROUND(SUM(Profit)/NULLIF(SUM(Revenue),0)*100,2)        AS ProfitMargin,
    SUM(TotalOrders)                                        AS TotalOrders
FROM (
    SELECT SalesTerritoryKey,'Internet' AS src,
           SUM(SalesAmount) AS Revenue, SUM(Profit) AS Profit,
           COUNT(DISTINCT SalesOrderNumber) AS TotalOrders
    FROM fact_internet_sales GROUP BY SalesTerritoryKey
    UNION ALL
    SELECT SalesTerritoryKey,'Reseller',
           SUM(SalesAmount), SUM(Profit),
           COUNT(DISTINCT SalesOrderNumber)
    FROM fact_reseller_sales GROUP BY SalesTerritoryKey
) x
JOIN dim_sales_territory t ON x.SalesTerritoryKey = t.SalesTerritoryKey
GROUP BY t.SalesTerritoryKey, t.SalesTerritoryRegion,
         t.SalesTerritoryCountry, t.SalesTerritoryGroup;
```

Key Insight

This view is why the territory table on Page 2 shows combined channel revenue and margin. Without it, a DAX measure combining two fact tables would be required. The CASE WHEN pivot is the SQL equivalent of a Power BI matrix with two measures.

vw_reseller_performance: Page 5 top-10 table

Joins four tables (fact, reseller, geography, territory) and uses DENSE_RANK() as a window function to pre-calculate revenue rank. This means the Page 5 top-10 resellers table requires only a filter RevenueRank <= 10, with no RANKX DAX needed.

```sql
DROP VIEW IF EXISTS vw_reseller_performance;
CREATE VIEW vw_reseller_performance AS
SELECT
    r.ResellerKey, r.ResellerSK, r.ResellerName, r.BusinessType,
    g.City, g.State, g.Country,
    t.SalesTerritoryRegion AS Territory,
    t.SalesTerritoryGroup  AS Region,
    COUNT(DISTINCT f.SalesOrderNumber)  AS TotalOrders,
    SUM(f.SalesAmount)                  AS TotalRevenue,
    SUM(f.Profit)                       AS TotalProfit,
    ROUND(SUM(f.Profit)/NULLIF(SUM(f.SalesAmount),0)*100,2) AS ProfitMargin,
    ROUND(AVG(f.SalesAmount),2)         AS AvgOrderValue,
    SUM(f.DiscountAmount)               AS TotalDiscount,
    DENSE_RANK() OVER (ORDER BY SUM(f.SalesAmount) DESC) AS RevenueRank
FROM fact_reseller_sales    f
JOIN dim_reseller            r ON f.ResellerKey       = r.ResellerKey
JOIN dim_geography           g ON r.GeographyKey      = g.GeographyKey
JOIN dim_sales_territory     t ON f.SalesTerritoryKey = t.SalesTerritoryKey
GROUP BY r.ResellerKey, r.ResellerSK, r.ResellerName, r.BusinessType,
         g.City, g.State, g.Country,
         t.SalesTerritoryRegion, t.SalesTerritoryGroup;
```

Key Insight

7 of the top 10 resellers by revenue have negative or near-zero profit margins. This is not visible from total revenue alone. The ProfitMargin column in this view, formatted with conditional coloring in Power BI, makes it immediately visible.

Analytical SQL Queries

Ten queries validate all Python EDA findings independently. Four are shown below, demonstrating the SQL techniques used in this project.

```sql
-- Q1: YoY Growth with LAG window function
WITH yr AS (
    SELECT Year, SUM(TotalRevenue) AS Revenue FROM vw_sales_summary GROUP BY Year
)
SELECT Year,
       ROUND(Revenue/1e6,2) AS RevenueMillion,
       ROUND((Revenue - LAG(Revenue) OVER (ORDER BY Year)) /
              NULLIF(LAG(Revenue) OVER (ORDER BY Year),0)*100,2) AS YoYGrowthPct
FROM yr ORDER BY Year;
```

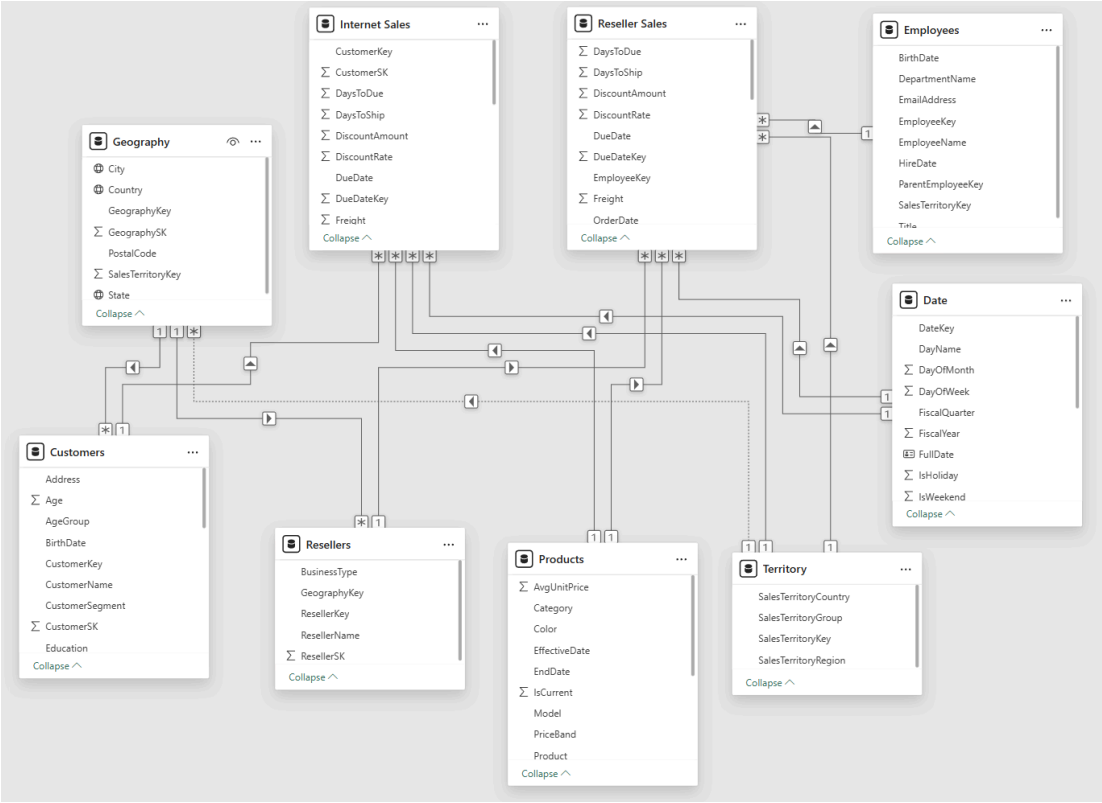
```sql
-- Q3: Territory Risk (high revenue, low margin)
SELECT Territory, Country,
       ROUND(TotalRevenue/1e6,2) AS RevenueMillion, ProfitMargin,
       CASE WHEN ProfitMargin < 20 AND TotalRevenue > 1e6 THEN 'Priority Fix'
            WHEN ProfitMargin < 30 THEN 'Watch'
            ELSE 'Healthy' END AS RiskFlag
FROM   vw_territory_performance ORDER BY TotalRevenue DESC;
```

```sql
-- Q9: Age group analysis
SELECT AgeGroup,
       COUNT(DISTINCT CustomerKey) AS Customers,
       ROUND(SUM(TotalRevenue)/1e6,2) AS RevenueMillion,
       ROUND(AVG(AvgOrderValue),2) AS AvgOrderValue
FROM   vw_customer_segments GROUP BY AgeGroup ORDER BY RevenueMillion DESC;
```

4. Power BI: Data Model, DAX Measures

Data Model

The Power BI model mirrors the MySQL star schema. 11 Active Many-to-One relationships renamed for convenience.



DAX Measures

> **Note: DATEADD vs SAMEPERIODLASTYEAR**
>
> Revenue YoY % uses DATEADD instead of SAMEPERIODLASTYEAR. SAMEPERIODLASTYEAR on an incompletely configured Date table returns values so close to zero they display as 2.0E-12 (scientific notation). DATEADD resolves correctly once the Date table is marked. All measures are stored in a single _Measures table for auditability.

Measure	DAX Formula	Output
Total Revenue	SUM(Internet[SalesAmount]) + SUM(Reseller[SalesAmount])	$109.8M

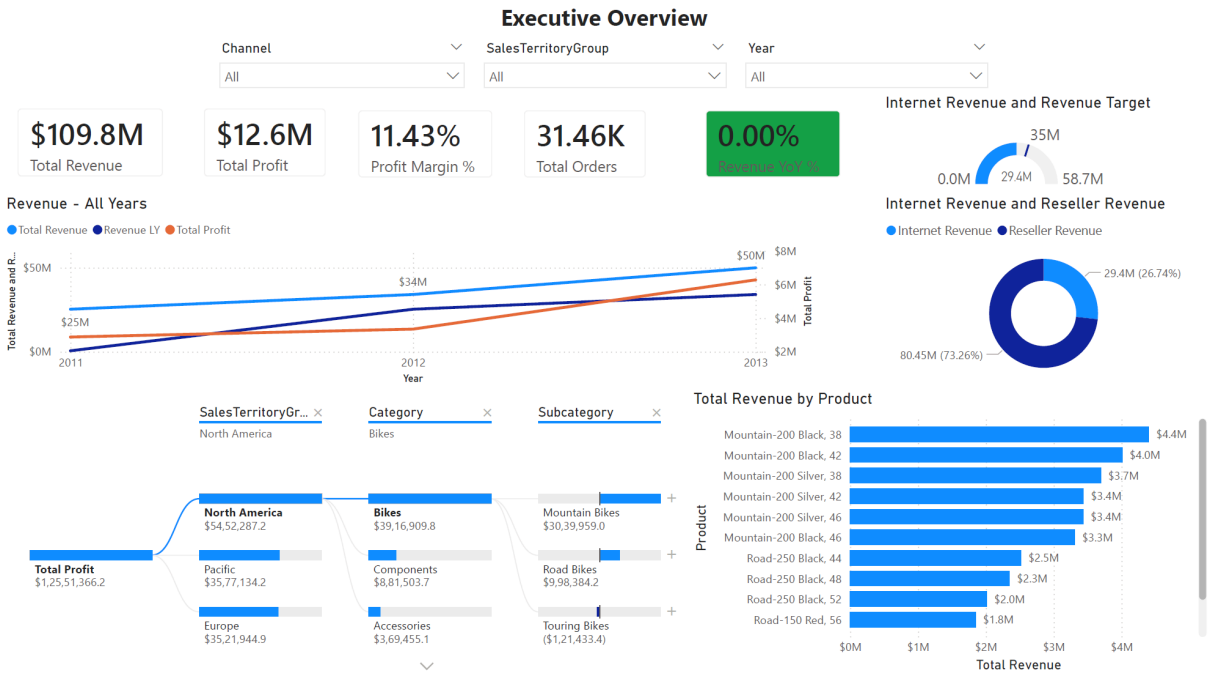
Internet Revenue	SUM('Internet Sales'[SalesAmount])	$29.4M
Reseller Revenue	SUM('Reseller Sales'[SalesAmount])	$80.5M
Total Profit	SUM(Internet[Profit]) + SUM(Reseller[Profit])	$12.6M
Profit Margin %	DIVIDE([Total Profit],[Total Revenue],0)*100	11.43%
Internet Margin %	DIVIDE(SUM(Internet[Profit]),[Internet Revenue],0)*100	41.15%
Reseller Margin %	DIVIDE(SUM(Reseller[Profit]),[Reseller Revenue],0)*100	0.58%
Total Orders	DISTINCTCOUNT(Internet[SalesOrderNumber]) + DISTINCTCOUNT(Reseller[SalesOrderNumber])	31,455
Unique Customers	DISTINCTCOUNT('Internet Sales'[CustomerKey])	18,484
Avg Order Value	DIVIDE([Total Revenue],[Total Orders],0)	$3,491
Revenue LY	CALCULATE([Total Revenue], DATEADD('Date'[FullDate],-1,YEAR))	Prior year value
Revenue YoY %	VAR ly=[Revenue LY] RETURN IF(ISBLANK(ly)||ly=0,BLANK(),DIVIDE([Total Revenue]-ly,ly,BLANK())*100)	2013: +46.7%
Territory Margin %	DIVIDE(SUM(Internet[Profit])+SUM(Reseller[Profit]), SUM(Internet[SalesAmount])+SUM(Reseller[SalesAmount]),0) *100	Per territory

North America Revenue	CALCULATE([Total Revenue],Territory[SalesTerritoryGroup]="North America")	$79.4M
Low Margin Territories	COUNTROWS(FILTER(VALUES(Territory[SalesTerritoryRegion]),[Territory Margin %]<10&&[Total Revenue]>1000000))	5
Product Rank	RANKX(ALL(Products),[Total Revenue],,DESC,Dense)	1–397
Revenue Target	35000000	$35M
Revenue vs Target %	DIVIDE([Internet Revenue],[Revenue Target],0)*100	84%

5. Dashboard Pages

→ Page 1: Executive Overview

“What is the total revenue picture and how is each channel performing?”



Revenue YoY% will show the value one a year is selected, otherwise the cumulative as shown is zero.The revenue trend (2011–2013) plots Total Revenue, Revenue LY, and Total Profit on dual axes. The Decomposition Tree allows drill-through from Total Profit through SalesTerritoryGroup and Category, immediately revealing North America Bikes as the profit driver. The channel donut (Reseller 73.3%, Internet 26.7%) and Internet Revenue gauge ($29.4M vs $35M target) frame the central question answered by Page 5.

Key Insight

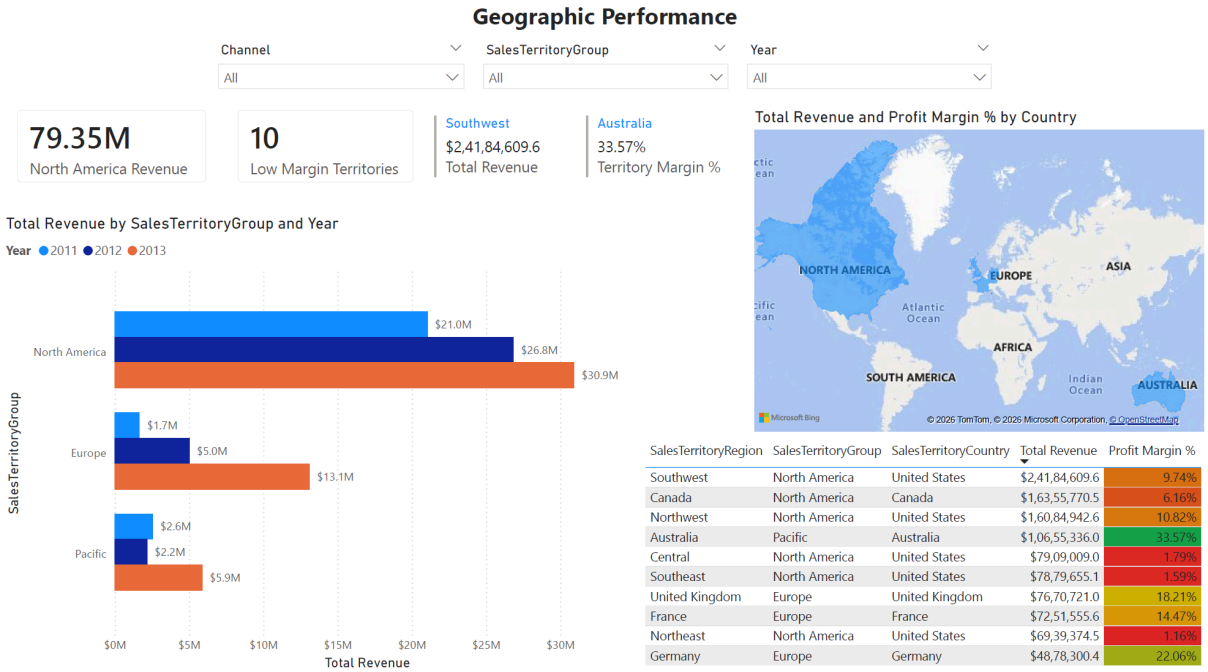
Revenue grew 34.7% in 2012 and 46.7% in 2013. The Internet channel tripled from $5.8M (2012) to $16.4M (2013). Strong growth trajectory confirmed across all years.

Business Action

The Revenue LY line shows the growth gap widening each year. The Decomp Tree drill confirms this is driven by North America Bikes. Maintain current product-channel investment while investigating the 2013 acceleration factor.

→ Page 2: Geographic Performance

“Which territories generate revenue and which are destroying margin?"



SalesTerritoryRegion	SalesTerritoryGroup	SalesTerritoryCountry	Total Revenue	Profit Margin %
Southwest	North America	United States	$2,41,84,609.6	9.74%
Canada	North America	Canada	$1,63,55,770.5	6.16%
Northwest	North America	United States	$1,60,84,942.6	10.82%
Australia	Pacific	Australia	$1,06,55,336.0	33.57%
Central	North America	United States	$79,09,009.0	1.79%
Southeast	North America	United States	$78,79,655.1	1.59%
United Kingdom	Europe	United Kingdom	$76,70,721.0	18.21%
France	Europe	France	$72,51,555.6	14.47%
Northeast	North America	United States	$69,39,374.5	1.16%
Germany	Europe	Germany	$48,78,300.4	22.06%

The territory table, sourced from vw_territory_performance (both channels pre-combined in MySQL), uses conditional formatting range on Profit Margin %: red below 10%, amber 10–25%, green above 25%. All five US regional territories are amber or red. Australia is the only green row. The stacked bar by Region and Year confirms all regions grew every year.

Key Insight

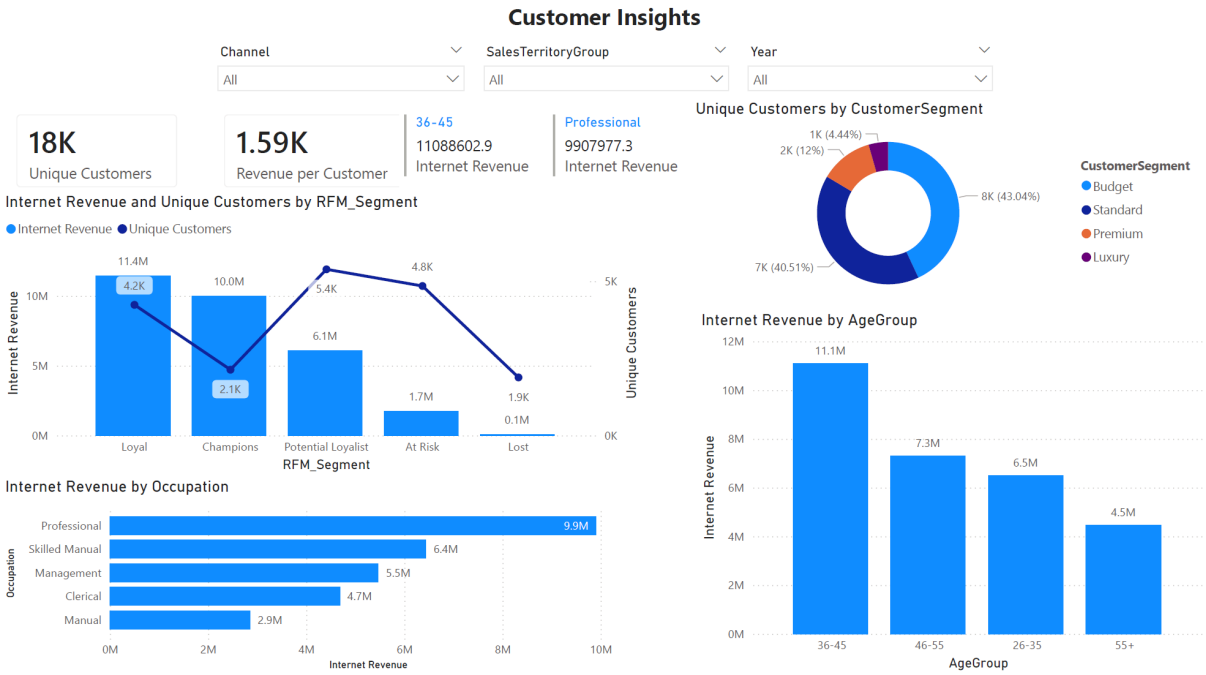
Geographic margin is determined by channel mix. Australia (33.6%) and UK (18.2%) outperform SW ($24.2M, 9.7%) because they are Internet-channel dominant. Reseller-heavy US territories carry Reseller margin (0.6%), not Internet margin (41.2%).

Business Action

Expand Internet channel in Australian and European markets before growing Reseller presence. Every new Internet order in those markets generates 41% margin vs 0.6% from a Reseller order.

→ Page 3: Customer Insights

“Who are the Internet customers and which segments need the most attention?”



All visuals are Internet Sales only. FactResellerSales has no CustomerKey by design; B2B transactions go through companies. The RFM combo chart (revenue bars + customer count line) shows Champions (2.1K) generating $10.0M and Lost (1.9K) generating $0.1M. The 18–25 age bar is near-zero: premium pricing naturally excludes this segment.

Key Insight

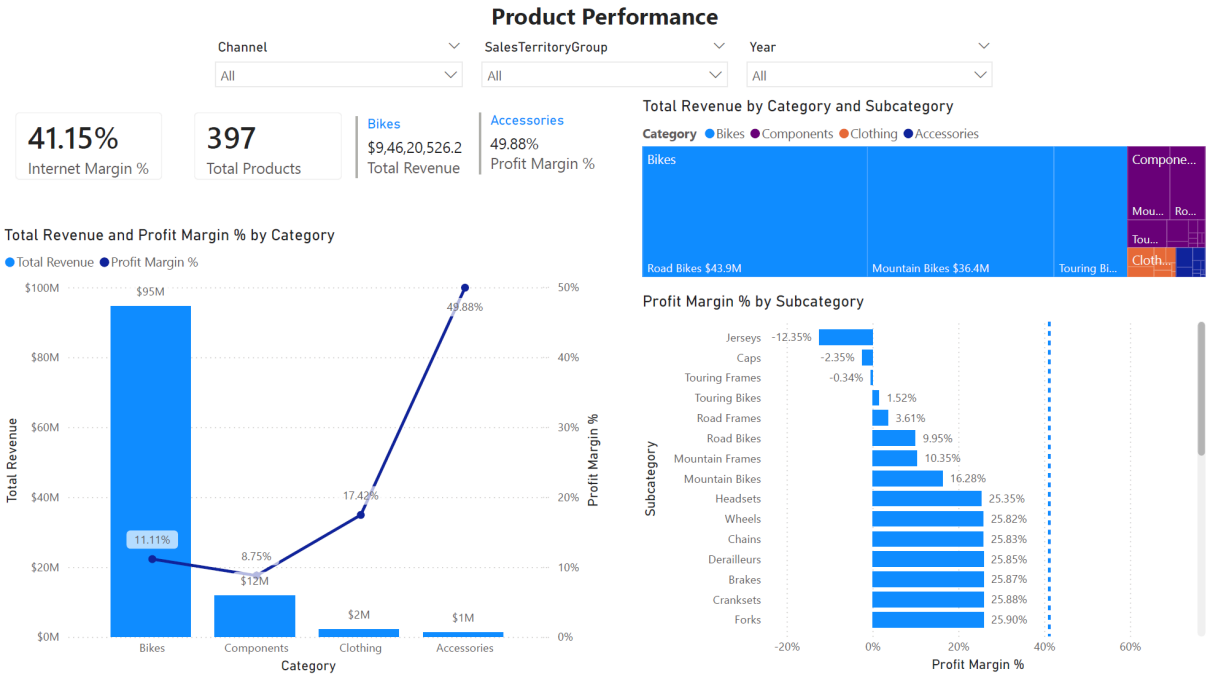
Professional + Management = 52.4% of Internet revenue. Budget customers are 43% of the base but a revenue minority. Champions generate 100× more revenue per customer than Lost segment customers of similar count.

Business Action

Activate VIP programme for 2,133 Champions. Launch re-engagement campaign for 4,846 At Risk customers (last purchase ~60 days ago). All marketing campaigns target Professional occupation, 35–55 age bracket.

→ Page 4: Product Performance

“Which categories drive revenue and where is margin opportunity hidden?”



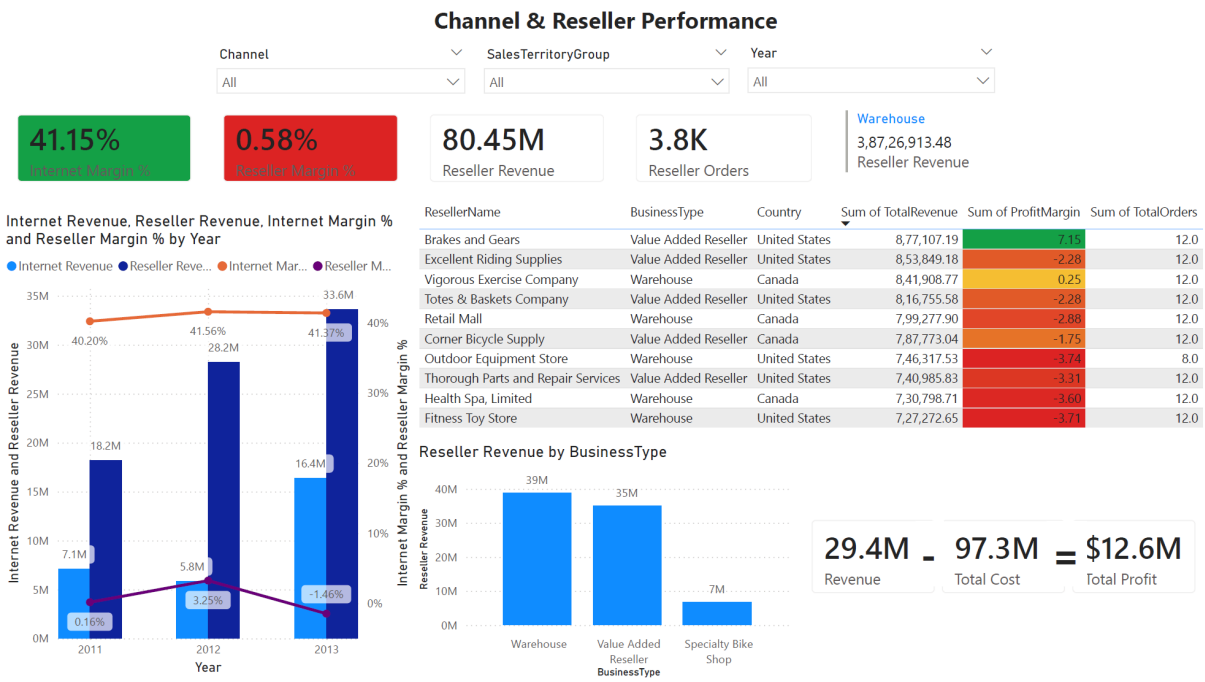
The combo chart (revenue columns + margin line) shows Bikes at $94.6M with 11.1% margin and Accessories at $1.3M with 49.9% margin. The visual contrast between the tallest bar and the highest margin line communicates the cross-sell opportunity in a single glance. The subcategory margin chart sorted ascending (worst first) immediately shows Jerseys and Touring Frames as loss-making.

> **Key Insight**
>
> Accessories carry 49.9% margin but only 1.3% of Bikes revenue. Every Bike customer is a natural Accessories buyer. The margin gap is not a product problem — it is a cross-sell execution gap.

> **Business Action**
>
> Implement Accessories bundle recommendation at checkout. A 10% attach rate on Bike purchases generates ~$9.5M additional high-margin revenue at zero acquisition cost. Reprice Jerseys (−12.4% margin) or discontinue.

→ Page 5: Channel & Reseller Performance

“Is the Reseller channel genuinely profitable or just high-volume?”



ResellerName	BusinessType	Country	Sum of TotalRevenue	Sum of ProfitMargin	Sum of TotalOrders
Brakes and Gears	Value Added Reseller	United States	8,77,107.19	7.15	12.0
Excellent Riding Supplies	Value Added Reseller	United States	8,53,849.18	-2.28	12.0
Vigorous Exercise Company	Warehouse	Canada	8,41,908.77	0.25	12.0
Totes & Baskets Company	Value Added Reseller	United States	8,16,755.58	-2.28	12.0
Retail Mall	Warehouse	Canada	7,99,277.90	-2.88	12.0
Corner Bicycle Supply	Value Added Reseller	Canada	7,87,773.04	-1.75	12.0
Outdoor Equipment Store	Warehouse	United States	7,46,317.53	-3.74	8.0
Thorough Parts and Repair Services	Value Added Reseller	United States	7,40,985.83	-3.31	12.0
Health Spa, Limited	Warehouse	Canada	7,30,798.71	-3.60	12.0
Fitness Toy Store	Warehouse	United States	7,27,272.65	-3.71	12.0

The 41.15% green and 0.58% red KPI cards side-by-side are the most impactful two numbers in the dashboard. The combo chart shows Reseller revenue growing to $33.6M (2013) while the margin line dips to −1.46%. The top-10 resellers table from vw_reseller_performance (filtered RevenueRank <= 10) with conditional formatting shows 7 of 10 at negative margins. The profit bridge (29.4M − 97.3M = $12.6M) shows Internet channel economics in oversized typography.

> **Key Insight**
>
> The Internet delivers 69× more profit per revenue dollar. Reseller revenue grew 85% from 2011–2013 while becoming loss-making at peak volume in 2013.

> **Business Action**
>
> Set a minimum 5% margin floor for all Reseller contracts. Exit the 7 top-10 resellers operating at negative margins. Reinvest recovered margin budget into Internet channel growth where every $1 of revenue generates $0.41 of profit.

6. Key Findings & Recommendations

Objective 1: Underperforming Territories

5 of 10 territories operate below 10% margin. Southwest ($24.2M) at 9.7%, Central at 1.8%, Northeast at 1.2%. Driven by Reseller channel dominance.

Recommendation: set minimum 5% margin floor for Reseller contracts. Pilot Internet channel investment in lowest-margin territories.

Objective 2: Revenue and Profitability Trends

Revenue grew 97.6% from 2011–2013. Blended margin 11.43% masks Internet (41.2%) vs Reseller (0.6%) divergence. Reseller went loss-making in 2013 at peak revenue.
Recommendation: report Internet and Reseller as separate P&L entities.

Objective 3: Customer Segmentation

Professional + Management = 52.4% of Internet revenue. Budget customers = 43% of base but minority of revenue.

Recommendation: allocate marketing budget to Professional/Management segments. Reduce investment in Manual/Clerical which show below-average spend.

Objective 4: Target Age Group

36–45 age group = $11.1M (37.8% of Internet). 18–25 near-zero.

Recommendation: all campaigns target 35–55. Do not attempt youth market penetration without pricing strategy changes.

Objective 5: Channel Comparison

Internet: $29.4M at 41.2% margin. Reseller: $80.5M at 0.6%. The Internet generates approximately 96% of total profit from 26.6% of revenue.

Recommendation: Internet growth and Reseller margin floor enforcement is the highest-ROI commercial strategy available.